THE GHOST IN THE MACHINE

Authorship and the Era-Detector Illusion

A five-stage investigation into how an AI-text detector recognised authorship, or whether it just learned the 19th-century literary style and treated that as “human”.

> The detector reached 100% accuracy and is still wrong about what it learned. Three independent methods (probes, saliency, adversarial search) all land on the same verdict.

KEYWORDS

authorship attribution · AI-text detection · shortcut learning · spurious correlation · stylometry · genre confound · register bias · LoRA · integrated gradients · adversarial evaluation · held-out probes · interpretability

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GENRE-MATCHED CLASSES

100%
IN-DISTRIBUTION ACCURACY

0.062
BEST ADVERSARIAL P(HUMAN), 8 GENERATIONS

PROLOGUE · SCOPE

What I did, what I didn't, and why

> *Before getting into results, here is what I actually built and what I skipped.*

All 5 tasks were completed end to end. The defining decision was to treat the genre/era confound as a controlled variable rather than a nuisance.

Task	Status	One-line result
0 · Dataset	Done	3 genre-matched classes (420 each), frozen split, 2 held-out probe sets.
1 · Stylometry	Done	Sentence-variance and punctuation separate classes; vocabulary barely does.
2 · Detectors	Done	XGBoost, FFNN, DistilBERT+LoRA, RoBERTa+LoRA, up to 100%.
3 · Interpretability	Done	Saliency and proper-noun ablation locate the signal: register, not names.
4 · Adversarial	Done	GA caps at P(human)=0.062; only archaic-word injection helps.

The decision that defined the submission

I did not want my human class and AI class to differ in authorship. If they did, the model would learn the genre, not the authorship. To deal with this I genre-matched the three primary classes, so the headline detector measures origin and not topic or era. I also used 2 probe diagnostic sets, modern prose and my own statement of purpose, never trained on, just used for prediction. Their whole job was to measure how much register still leaks.

CENTRAL THESIS

The detector never learned to recognise AI. It learned 19th-century literary register and used that as a stand-in for human writing. I prove this three independent ways.

Honestly scoped out

- Two authors only (Austen and Dickens, my favourites). Enough for a clean authorship contrast on a free compute budget.
- LoRA on DistilBERT and RoBERTa, not full fine-tunes of larger models.
- Topic modelling was dropped as an automatic step and replaced with hand-curation. More on why in Chapter 0.

Building the dataset

> *These are closer to lab notes than prose. The data is where everything later either holds up or falls apart, so I want the messy version on record.*

Goal: three genre-matched classes, 420 paragraphs each, plus two held-out probe sets. Remove the era confound at the source, then measure whatever leaks through anyway.

Class 0 · Human
Essayistic passages from Austen and Dickens.

Class 1 · AI-neutral
LLM essays, modern register.

Class 2 · AI-styled
LLM essays imitating the author.

Probe A: 60 paragraphs of modern human prose. Probe B: 11 paragraphs from my own SOP.

The core move: genre matching

I kept only the essayistic human paragraphs and threw out the dialogue-heavy and name-dense narrative ones. That way all three classes share register, which forces the detector toward authorship. The probes then measure what is left over.

What actually happened, in order

- First paragraph splitter had an off-by-one that ate the last line of every chapter. Caught it late, after I had already run one round of stylometry on the broken set and the numbers looked weirdly clean. Re-ran everything.
- Wrote a tiny throwaway script just to print word-count histograms per class. Ugly, but it is how I noticed the length gap I mention below.
- I kept forgetting to log the generation prompts at first, so my early runs were not reproducible. Added JSONL logging after losing one batch I could not reconstruct.

SETBACK 1 · TOPIC MODELLING

LDA on 400 short fiction paragraphs gave me character names and filler (“mr”, “elizabeth”, “joe”), not themes. I added a custom stop-word list and bigrams. Names receded but the topics stayed as loose word-bags (“dinner, hours, half, years”).

DECISION

I stopped torturing the tuning and treated LDA as a hypothesis generator. I hand-curated 10 topic labels grounded in themes that kept recurring across runs: family, marriage, fear, character, social standing. The fact that they recurred is the evidence they are real.

SETBACK 2 · GENERATION PROVIDER

Gemini 2.5 Flash kept returning 503 / RESOURCE_EXHAUSTED and hit a hard daily cap. Even with backoff and a tighter length prompt, a lot of calls came back empty and got discarded. I added `print(repr(raw))` and saw the responses were just empty, so it was the API, not my length filter.

DECISION

Switched to Groq, Llama-3.3-70B. Reliable and faster. Every call is logged so it is auditable, and length is enforced in the prompt and re-checked after generation.

SETBACK 3 · AUTHOR IMBALANCE

First human pull came out roughly 2:1 Austen to Dickens. Dickens is more dialogue-heavy, so fewer of his paragraphs survive the essayistic filter. Small finding in itself.

DECISION

Added a third Dickens book, Oliver Twist. That rebalanced the corpus to 216 Austen, 208 Dickens.

A length signal I left in on purpose

Mean word counts differ by class: Human 153.9 ± 44.8, AI-neutral 166.7 ± 12.2, AI-styled 145.4 ± 8.9. I did not scrub this. Task 2's leakage audit is built to catch exactly this kind of thing, so leaving it in keeps the later numbers honest.

Stylometry

> *The classes separate before any model is trained. So Task 2 becomes a simpler question: can a model recover what plain statistics already show?*

Feature	Human	AI-neutral	AI-styled	Reading
Sentence-length variance	18.1	5.1	6.9	Humans vary rhythm ~3x
Semicolons / 1k words	15.0	0.0	0.1	Near-binary tell
Em-dashes / 1k words	5.9	0.0	0.0	AI uses none
Hapax (fixed-window)	0.549	0.529	0.502	Barely separates

What stood out

Sentence-length variance is the strongest separator. AI prose is rhythmically flat. Humans swing between long and short sentences a lot more. Punctuation is a strong tell too, with one caveat: the heavy semicolon and em-dash use is partly an era signal, since 19th-century authors punctuate heavily. Lexical diversity barely separates the classes at all. The signal lives in structure and punctuation, not vocabulary.

FORESHADOWING

SYNTHESIS · PCA

Projected onto two components, the AI classes collapse into a tight, low-variance band while the human prose sprawls. Neutral and styled overlap heavily. That overlap quietly foreshadows that neutral-vs-styled is the hard problem in Task 2.

TTR and Hapax are both computed on fixed-length samples, since both depend on length.

Three tiers, one frozen split

This chapter stays formal on purpose. The detectors are set up as an ablation, not a leaderboard.

All detectors run on the same frozen split, and a leakage audit was done before anything else.

Leakage audit

Zero text overlap across splits. But a length-only classifier reached 0.71, so length is a partial confound. I report it openly. The real detectors lean on punctuation and rhythm, not just length.

Detector	Human-vs-AI	Neutral-vs-Styled	Tertiary
Tier A · XGBoost (stylometry)	0.968	0.802	0.841
Tier B · FFNN (mean GloVe)	0.995	n/a	0.995
Tier C · DistilBERT + LoRA	1.000	n/a	1.000
Tier C · RoBERTa + LoRA	1.000	1.000	1.000

In Tier A, punctuation dominates the feature importances (semicolon, em-dash), which matches Chapter 1. The hard neutral-vs-styled framing drops to 0.80, confirming the two AI classes sit close together.

RED HERRING

Tier B's 0.995 is misleading. It is partly real generalisation, partly an era confound hiding in the embeddings. On the modern-human probe it still labels about 85% as human, so some of that score is genuine and some is just the model reading era off the vocabulary.

SETBACK · SUSPICIOUSLY PERFECT TRANSFORMERS

DistilBERT and RoBERTa both hit 100% on the test set. That is a reason to dig in, not to celebrate. In-distribution accuracy is saturated and cannot tell the two models apart.

DECISION

Let the probes do the discriminating instead. One parameterised loop trains both models and evaluates each on the held-out probes. The table below is the real comparison.

Model	Test acc	Modern-human, %human	SOP, %human
DistilBERT	1.000	70.0%	0.0%
RoBERTa	1.000	86.7%	0.0%

Same test accuracy, very different probe behaviour. RoBERTa generalises to modern human prose much better (86.7% vs 70%), so I made it the canonical model for Tasks 3 and 4. Both still call the AI-assisted SOP 0% human, which is consistent across architectures.

Interpretability

> *Four experiments, all asking what the detector actually keys on. They agree.*

Saliency

Styled-AI imposters get flagged on Victorian-pastiche words (`season`, `affection`, `love`, `truth`, the "it was the season of..." cadence). AI-neutral gets flagged on modern abstract vocabulary (`shaping`, `environment`, `strategies`, `framework`, `transition`). Two AI classes, two different cue-sets.

Proper-noun ablation

DELTA = 0.000

Masking every PERSON, GPE and ORG entity changed accuracy by nothing at all. The model relies on none of them.

SOP disentangling

The SOP is AI-assisted and also modern in register, so its "AI" label is entangled. Saliency separates the two. The top tokens are modern domain vocabulary (`AI`, `systems`, `research`, `models`, `researcher`) with zero LLM-cliché words ("delve", "leverage", "underscore"). So the SOP is flagged for modern register, not for any detectable generative fingerprint.

Genre probe

Probe set	Mean P(human)	Reading
Modern human essays	0.825	Genuine human prose, mostly correct
My SOP (AI-assisted)	0.000	Modern register read as AI

Genuine modern human prose scores 82.5% human. My SOP scores 0%. That gap is the era-vs-AI signal, partly separated. The cleanest evidence of the confound lives in the modern-human probe, with the SOP honestly flagged as entangled.

Adversarial evolution

> *The foreshadowing from Chapter 1 pays off here. If the detector learned era, then to beat it you would have to sound old. Watch what the algorithm finds on its own.*

A genetic algorithm tries to evolve an AI paragraph until RoBERTa calls it human.

BUG I CAUGHT LATE

My first fitness function was reading the wrong softmax index, so it was rewarding P(AI) instead of P(human). Every "improving" individual was getting more obviously AI, not less. I only noticed because the best paragraphs looked worse each generation. Fixed the index and re-ran.

Generation	0	1 to 3	4	5	6	7
Max P(human)	0.000	0.000	0.000	0.001	0.006	0.062

RESULT

After 8 generations of pressure, the best it reached was 0.062. Never close to the 0.5 needed to flip the label. The only mutation that moved the needle at all was archaic-word injection.

The winning individual just piled on antiquated vocabulary: `thraldom`, `crepuscular`, `tenebrosity`, `vicissitudes`, `languor`, `lassitude`, `anhedonia`. The GA found, on its own, that the route to "human" is to sound like a 19th-century author. That adversarially confirms the era confound. The detector rewards Victorian register, not authorship.

THE LOOP CLOSES

Three independent methods (held-out probes, saliency, adversarial search) all land on the same conclusion: the detector learned era, not authorship.

Limitations, reproducibility, and what comes next

Where I think this is weak, and what I would try next if I kept going.

Limitations

- Two authors, one language, one era of human text. The human class is intrinsically Victorian, which is the exact confound this report is about.
- A length signal and an era-correlated punctuation signal both remain in the data. I measure them rather than removing them.
- The adversarial ceiling is compute-bounded at 8 generations, with a single mutator model that sometimes returns commentary instead of a clean rewrite.

What a v2 would add

- An era-balanced human class (modern human prose inside training) to test whether the confound dissolves.
- A punctuation- and length-ablated detector, to isolate authorship from register.
- A stronger adversarial operator and more generations, to find the real robustness ceiling.

Reproducibility

- One frozen stratified split (882 / 189 / 189) reused by every tier. Seeds fixed at 42.
- Every generation call logged to JSONL. Topics and the provider switch are documented.
- Notebooks run top to bottom with outputs preserved. LoRA adapters saved.

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CLOSING

What I am happiest about here is not the 100% accuracy. It is that I did not take it at face value. The number that actually matters is 0.062, because that is where the detector stopped being fooled and showed what it was really doing.